LECTURE NOTES ON DATA STRUCTURES

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1. Stacks

A stack is a linear data structure that follows the LIFO principle, where the last element inserted is removed first.

The time to push n elements is T = n * c where c is the cost of one push.

Kinetic energy is KE = 1/2 mv^2 where m is the mass.

```
#include <stdio.h>
int main() {
    int a = 5;
    printf("%d", a);
    return 0;
}
```

Push adds an element to the top of the stack and pop removes it.

2. Queues

A queue is a linear data structure that follows the FIFO principle in which the first element is removed first. Velocity is v = u + at where u is the initial velocity and a is acceleration.